Linear Algebra

Linear algebra is the branch of maths thats deal with vector, vector spaces, linear transformation, and system of linear equations. It provides a framework to understand the prop and operations of these mathematical obj, which can be represented using matrices and vectors.

- It is an foundational concept for ML, DL, NLP, CV
- Scalars, vectors, matrices, mathematical operation on matrices, Eigen value and vector.

→ **Applications:**

① Data representation and manipulation

Dataset: Create model which will be able to predict

- Vectors can ~~be multidimensional depending upon the parameters are added.~~ three features

→ Like House price predictor

Area	Rooms	Location	These are independent or input features	Price	Output or dependent features
1000	4	Mumbai		1.5 cr	

→ These mathematical values are stored in vectors, even the text converted to numbers and then stored.

→ After that, mathematical operations it quantifies the relationship b/w parameters. And give us the output.

• Quantify the relationship means to measure or express how strongly two or more variable are related using numbers.

[Here correlation is used which is part of stats]

→ The point is these relations are build just to make the model think that which parameter will effect the output by how much.

→ As our human brains are trained by birth that we know if any of the parameter is increase, it will effect output.

→ So here linear algebra represent data in vectors and matrices, and manipulate them by doing mathematical operations to analyze and extract some insights.

(*) Linear algebra works well with higher dimensional data. By dimensional reduction.
↳ PCA

Like 500 dimensions / features → 2/3 dimensions
so we can understand.

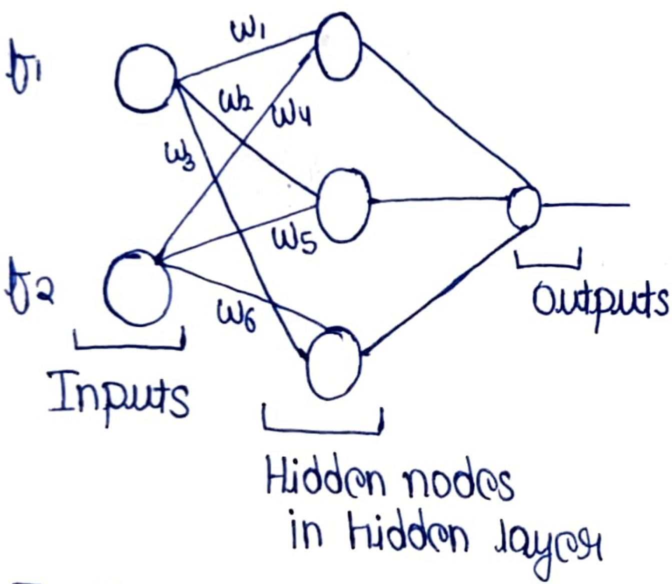
→ Vectors become multidimensional as more features (parameters) are added to a dataset. Each feature represent one dimension of the vector.

② Machine Learning and Artificial Intelligence

i) Model Train : On Linear Algebra → By Matrix multiplication
↓
or can say matrix arithmetic operations

ii) Dimensionality reduction : By PCA → In Linear Algebra → By Eigen value and Eigen vector.
↳ Reduce from higher dimension to lower dimension

iii) Neural Networks: Forward propagation and Backward propagation.



f_1 f_2
Area No. of rooms

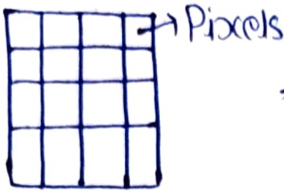
$$\begin{bmatrix} f_1 \\ f_2 \end{bmatrix} \begin{bmatrix} w_1 & w_2 & w_3 \\ w_4 & w_5 & w_6 \end{bmatrix} \Rightarrow \text{matrix multiplication}$$

Why GPUs ~~str~~ perform faster.

- GPU → Have cores → where matrix multiplication done parallely.
- $\underbrace{\text{Tensor Flow}}_{\text{Libraries}}$ ⟶ Convert all matrix value to tensors

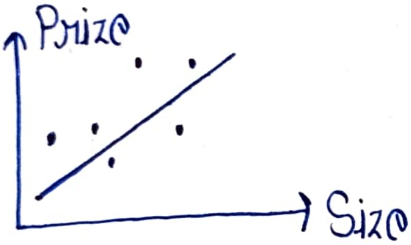
③ Computer Graphics:

Image
(RGB)



→ Scaling, Rotating,
Black and white
↳ This all involves
Linear Algebra
↳ Because of
Transformations.

④ Optimization:



• We Here solve equations
like linear equations.

$y = mx + c$ → Regression Algorithm

Slope (m), Intercept (c)

→ The line is not drawn
random, its solved and
the values we get is
the maximum of f^n and where
errors are minimal.